

PROJECT SUMMARY

This project implements a fully functional 24-hour digital alarm clock in VHDL for the Nexys A7-50T FPGA board. The system displays the current time and the stored alarm time, supports interactive time and alarm configuration, and provides both visual blinking and an audible piezo alert when the alarm is triggered.

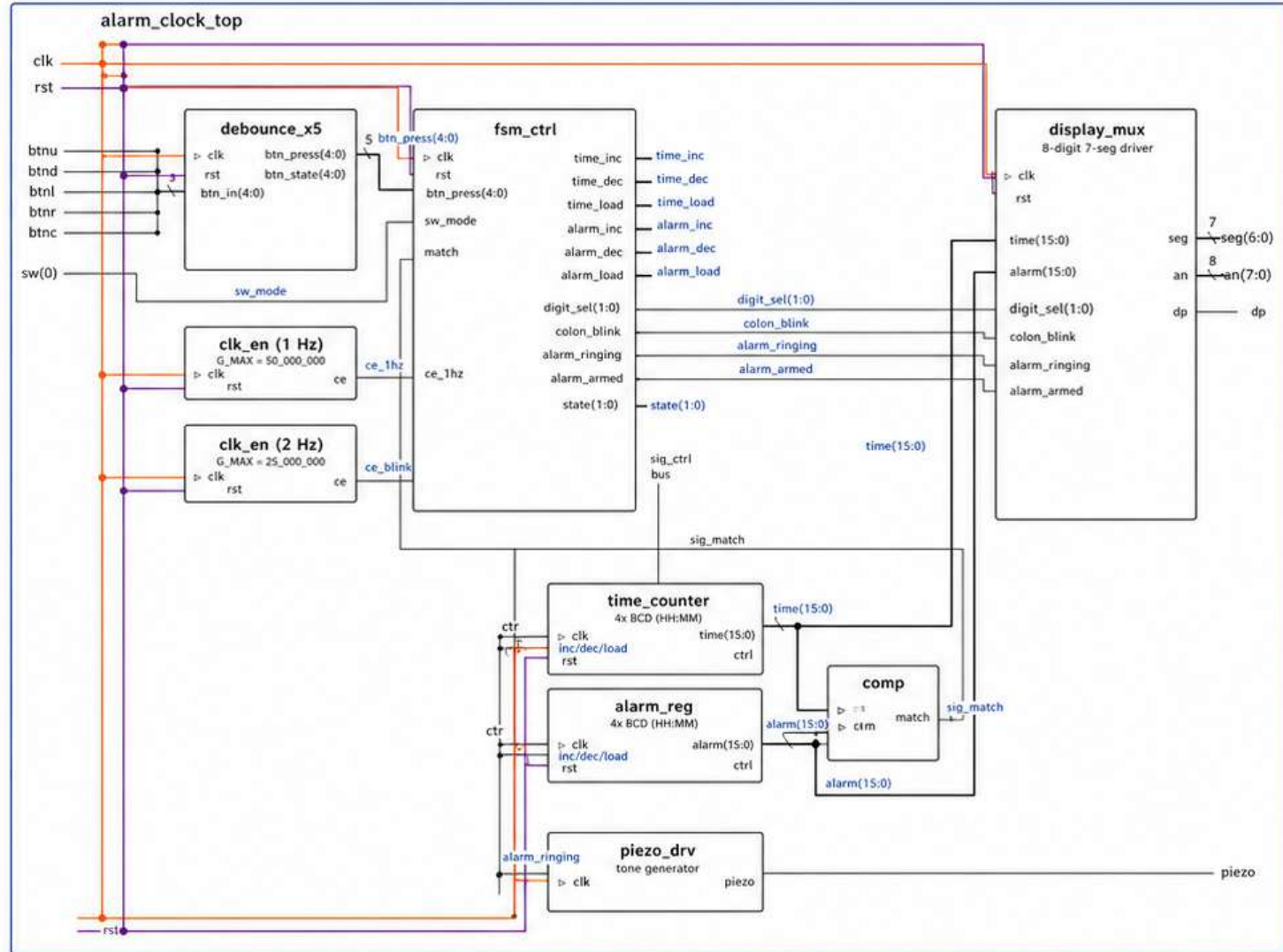
KEY FEATURES

- SET_TIME**
Activated by switch SW(0). LEFT and RIGHT select the active digit, UP and DOWN change its value. Returning SW(0) to the default position starts the clock again.
- SET_ALARM**
In RUN mode, press BTNC to edit the alarm. LEFT and RIGHT select a digit, UP and DOWN change it. Press BTNC again to save and arm the alarm.
- ALARM_RING**
When the current time matches the stored alarm time, the piezo buzzer sounds and the right display blinks. Press BTNC to stop the alarm.

MODULES

debounce_x5	Filters push-button bouncing for stable inputs.
clk_en (1 Hz)	Generates a 1 Hz enable pulse for real-time counting.
clk_en (2 Hz)	Generates a 2 Hz enable pulse for blinking indicators.
fsm_ctrl	Controls the RUN, SET_TIME, SET_ALARM and ALARM_RING states.
time_counter	Stores and edits the current time in BCD HH:MM.
alarm_reg	Stores the programmed alarm time in BCD HH:MM.
comp	Compares current time and alarm time.
display_mux	Drives the 8-digit seven-segment display.
piezo_drv	Generates the warning tone for the piezo buzzer.

BLOCK DIAGRAM



System block diagram.

